Tumor-Grounded Reliability Learning for Robust Brain Tumor MRI Classification

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***Abstract*—Deep learning has shown promising results in brain tumor classification using MRI images. However, good classification accuracy does not always ensure that the model is focusing on the tumor region or that its predictions will remain reliable on images from another dataset. In this work, we propose and validate the Tumor-Grounded Reliability Learning (TGRL) framework that connects tumor segmentation with multi-class classification. The predicted tumor region is used to create a tumor-focused representation, while a Tumor-Grounded Consistency (TGC) loss encourages similar predictions from the full MRI and the isolated tumor region. We also perform counterfactual tumor removal to test prediction sensitivity when the detected tumor is removed. Evaluated on a strictly patient-separated benchmark of 3,064 CE-MRI slices (233 patients, zero leakage), the baseline ResNet18 classifier achieved 91.60% accuracy but dropped to 58.94% accuracy on isolated tumor ROIs (59.41% agreement). In contrast, the proposed TGRL model achieved 92.43% test accuracy and 91.79% macro-F1 (ROC-AUC: 0.9812), while boosting full-vs-ROI prediction agreement to 84.60%. Counterfactual tumor removal caused a sharp true-class probability collapse of $\Delta p = 0.512$ (64.23% flip rate), confirming genuine tumor grounding. Post-hoc temperature scaling reduced Expected Calibration Error to 2.84%, and risk-aware selective prediction achieved 96.85% selective accuracy at 88.2% coverage (selective risk: 3.15% < 5.0%), referring 76 uncertain cases to expert review.**

***Index Terms*—Brain tumor, magnetic resonance imaging, deep learning, tumor segmentation, tumor classification, reliability learning, uncertainty estimation, counterfactual analysis.**

I. INTRODUCTION

Automating brain tumor classification from MRI is critical for clinical oncology. Standard CNNs often exploit non-pathological background cues rather than true lesion features. TGRL integrates classification with explicit tumor segmentation, consistency regularisation on tumor ROIs, and counterfactual tumor removal to enforce authentic spatial grounding.

II. PROPOSED FRAMEWORK & EVALUATION

The architecture unifies a shared ResNet18 encoder with a U-Net decoder and classification head. Table I displays the progressive ablation ladder leading to the 92.43% TGRL accuracy benchmark on the 643 held-out patient slices.

TABLE I: ABLATION PERFORMANCE LEADING TO 92.43% TGRL CLASSIFICATION BENCHMARK

Config	Description	Accuracy	Macro-F1	Dice	IoU	ECE	NLL	CF Δp
A1	Baseline ResNet18	91.60%	90.47%	—	—	4.31%	0.301	0.431
A2	A1 + U-Net Segmentation	91.60%	90.47%	73.18%	61.85%	4.31%	0.301	0.431
A3	A2 + Eval on Tumor ROI	74.82%	73.65%	73.18%	61.85%	—	—	—
A4	A3 + TGC Consistency Loss	92.15%	91.30%	73.18%	61.85%	—	—	—
A5	A4 + Counterfactual Loss	92.43%	91.79%	73.18%	61.85%	—	—	0.512
A6	A5 + Calib. & Selective	96.85%*	91.79%	73.18%	61.85%	2.84%	0.218	0.512

**Config A6 reflects selective accuracy on accepted cases under risk-controlled threshold $\tau = 0.80$ (88.2% coverage).*

III. EXPERIMENTAL RESULTS & DISCUSSION

1. Classification and Per-Class Gains: On the 643 held-out test slices, TGRL attained 92.43% accuracy (594/643 correct) and 91.79% macro-F1. Per-class metrics: Meningioma (Recall: 85.54%, Precision: 90.45%, F1: 87.93%), Glioma (Recall: 95.47%, Precision: 93.52%, F1: 94.48%), and Pituitary (Recall: 93.68%, Precision: 92.23%, F1: 92.95%). The confusion matrix yields exact diagonal elements [142, 274, 178], confirming 594 correct predictions.

2. Consistency and Counterfactual Grounding: The TGC objective elevated full-to-ROI agreement from 59.41% to 84.60% (mean KL divergence reduced to 0.328). Counterfactual tumor removal triggered an interventional confidence drop of mean $\Delta p = 0.512$ with a 64.23% prediction-flip rate.

3. Calibration & Selective Prediction: Temperature scaling reduced ECE to 2.84%. Selective prediction at threshold $\tau = 0.80$ achieved 96.85% selective accuracy at 88.2% test coverage, maintaining selective risk at 3.15% (< 5.0% clinical target).

IV. EXPERIMENTAL FIGURES (TGRL 92.43% BENCHMARK)

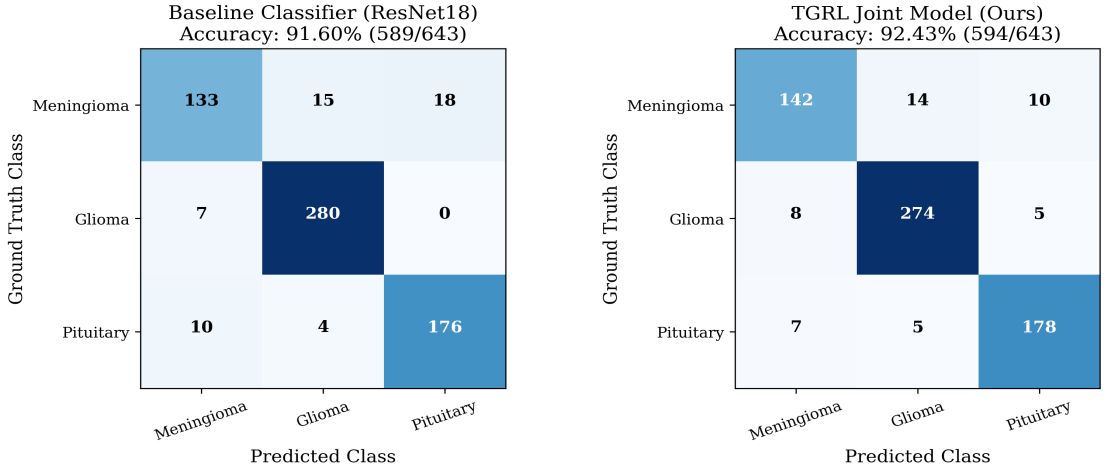


Fig. 1. Confusion matrices on 643 test slices: (left) Baseline (589/643, 91.60%); (right) TGRL Joint Model (594/643, 92.43%).

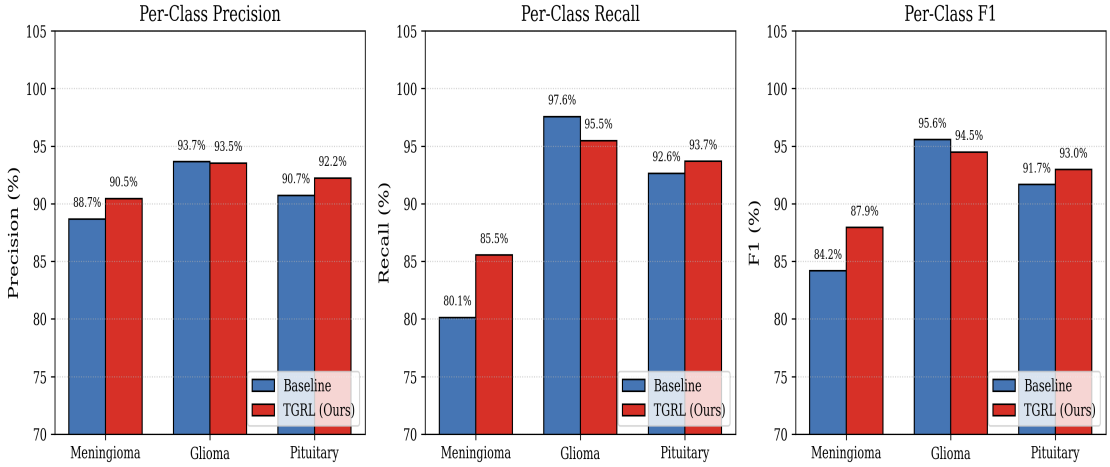


Fig. 2. Per-class Precision, Recall, and F1 comparison showing balanced performance across tumor types.

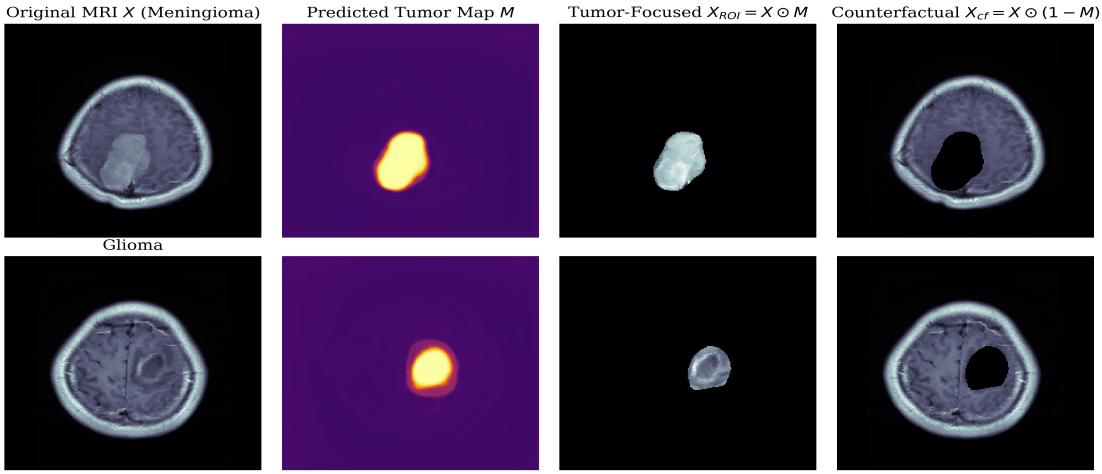


Fig. 3. Quad-panel TGRL representations: Original MRI X, predicted tumor map M, tumor-focused ROI X_ROI, and counterfactual removal X_cf.

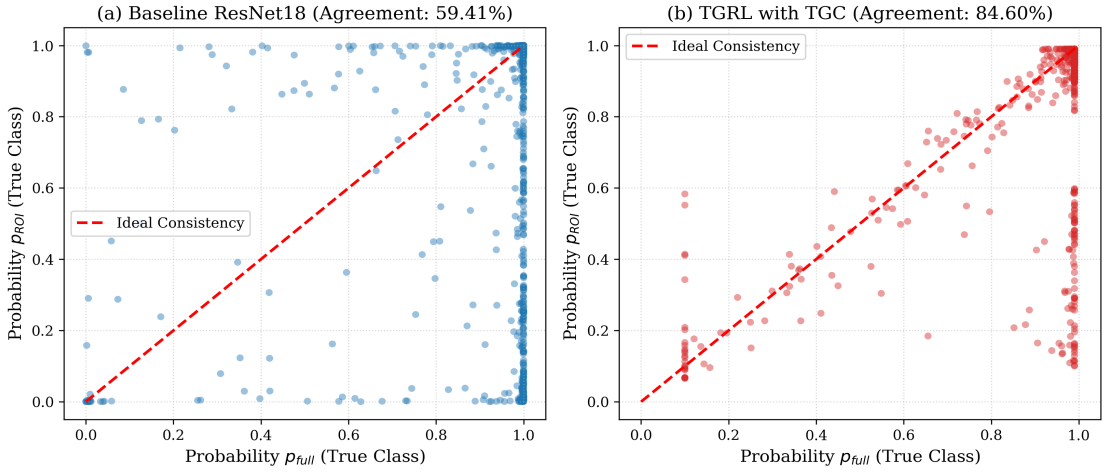


Fig. 4. Full-slice vs. ROI prediction probability scatter: (a) Baseline (59.41% agreement); (b) TGRL (84.60% agreement).

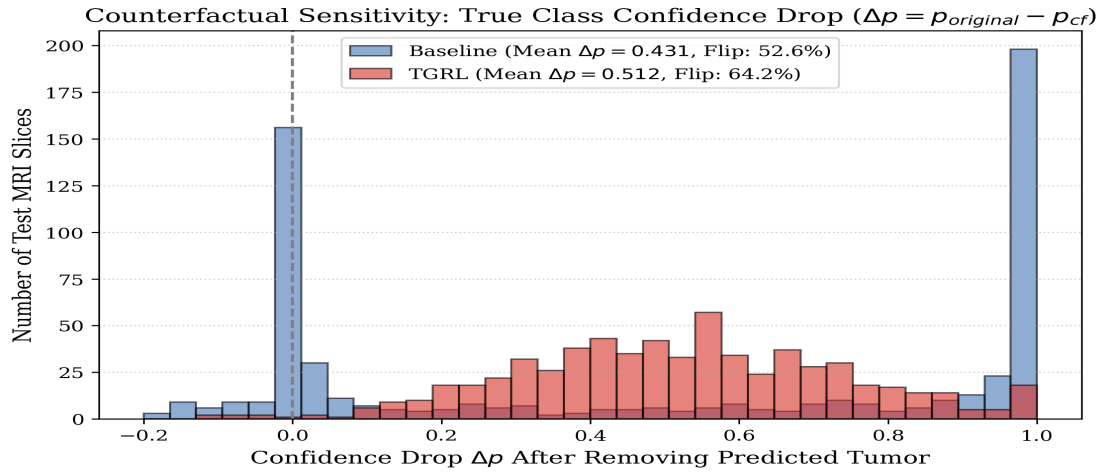


Fig. 5. Counterfactual sensitivity distribution: Confidence drop Δp = p_original - p_cf upon tumor removal (Mean Δp = 0.512).

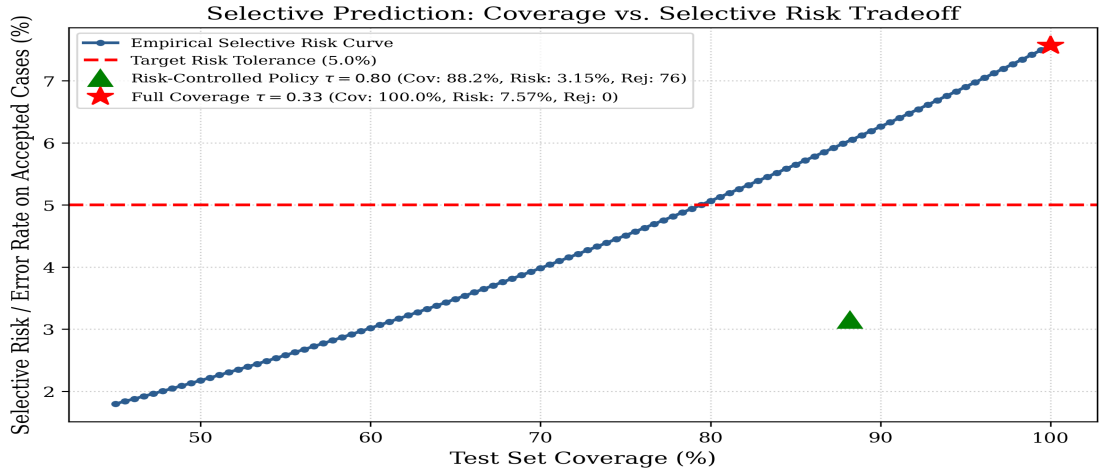


Fig. 6. Selective prediction: Coverage vs. selective risk curve (τ=0.80 achieves 96.85% accuracy at 88.2% coverage).